

Math 141, S10

Quiz 10

Section:

Name: Answer

SSN: xxx-xx-

Instructions: There is totally 1 problem. You must show all of your work to receive a credit for a correct answer. No calculator is allowed in the quiz and exam.

Problem 1.(10 points) Find two positive numbers whose product is 81 and whose sum is a minimum.

Solution.

$$Q = x + y$$

with $x \cdot y = 81$, and $x > 0, y > 0$.

$$x \cdot y = 81 \Rightarrow y = \frac{81}{x}$$

$$\text{Thus } Q(x) = x + \frac{81}{x}$$

with $x > 0$.

To minimize $Q(x)$, we find.

$$Q'(x) = 1 - \frac{81}{x^2} = 0$$

$$\Rightarrow x = 9 \rightarrow \text{critical number}$$

$$\text{When } 0 < x < 9, Q'(x) < 0 \Rightarrow Q(9) = 9 + \frac{81}{9} = 18$$

when $x > 9$, $Q'(x) > 0$ is an absolute minimum.

Thus, these two positive numbers are

$$x = 9, y = 9.$$